

100% Excellent!

PHY201 Exam 5 on chapter(s): 11, 12

Name: _____

These questions assess your understanding of the material from Chapter(s) 11, 12 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. Calculate the maximum buoyant force that saltwater could exert on a ship with a mass of 2.3×10^8 kg if the ship displaces 4.0×10^6 m³ of water. Saltwater has a density of 1025 kg/m³.

$$F_b = W_w$$

$$F_b = m_w g$$

$$F_b = \rho_w V g = 1025 \frac{\text{kg}}{\text{m}^3} \cdot 4 \times 10^6 \text{ m}^3 \cdot 9.8 \frac{\text{m}}{\text{s}^2} = 4.018 \times 10^{10} \text{ N}$$

ANSWER: $4.018 \times 10^{10} \text{ N}$ ✓

2. The aorta pumps blood from the heart to the rest of the body. The flow rate of blood through Ashley's aorta is 19.8 L/min and Ashley's aorta has a radius of 14 mm. Recall that blood flows through smaller blood vessels known as capillaries, which have an average diameter of 9.2 μm. The speed of blood in the capillaries is about 0.32 mm/s. Calculate the number of capillaries in Ashley's blood circulatory system.

$$Q = 19.8 \frac{\text{L}}{\text{min}} = \left| \frac{\text{m}^3}{1000 \text{ L}} \right| \left| \frac{\text{min}}{60 \text{ s}} \right| = 3.3 \times 10^{-4} \frac{\text{m}^3}{\text{s}}$$

$$r_c = \frac{d_c}{2} = \frac{9.2 \times 10^{-6} \text{ m}}{2} = 4.6 \times 10^{-6} \text{ m}$$

ANSWER: 1.55×10^{10} capillaries ✓

$$Q = A_1 v_1 = A_2 v_2$$

$$v = 0.32 \frac{\text{mm}}{\text{s}} \left(\frac{\text{m}}{1000 \text{ mm}} \right) = 0.32 \times 10^{-3} \frac{\text{m}}{\text{s}}$$

$$n_2 = \frac{Q}{A_2 v_2}$$

$$n_2 = \frac{Q}{\pi r_c^2 v_2} = \frac{3.3 \times 10^{-4} \frac{\text{m}^3}{\text{s}}}{\pi (4.6 \times 10^{-6} \text{ m})^2 \cdot 0.32 \times 10^{-3} \frac{\text{m}}{\text{s}}} = 1.551 \times 10^{10} \text{ capillaries}$$

3. Mark puts a coin on a scale and finds its mass is 5.7 g. When the coin and scale are completely submerged in freshwater, the scale reports the coins mass as 1.96 g. What is the density of the coin?

$$m = 5.7 \text{ g}$$

ANSWER: $1.524 \times 10^3 \frac{\text{kg}}{\text{m}^3}$ ✓

$$\sum \vec{F}_y = 0$$

$$W = F_b + N$$

$$mg = W_w + m_2 g$$

$$mg = \rho_w V g + m_2 g$$

$$m = \rho_w V + m_2, \quad V = \frac{m}{\rho_c}$$

$$\frac{m - m_2}{\rho_w} = \frac{m}{\rho_c}$$

$$\rho_c = \frac{\rho_w(m)}{m - m_2} = \frac{1000 \frac{\text{kg}}{\text{m}^3} \cdot 5.7 \text{ g}}{5.7 \text{ g} - 1.96 \text{ g}} = 1.524 \times 10^3 \frac{\text{kg}}{\text{m}^3}$$

+3

4. Jonathan attaches a nozzle of diameter 6.9 mm to a garden hose of diameter 2.0 cm. If the hose fills up a 3-L bucket in 11.7 seconds, calculate the pressure in the hose assuming level, frictionless flow.

$$P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2$$

$$Q = \frac{3L}{11.7s} \left(\frac{m^3}{1000L} \right) = 2.564 \times 10^{-4} \frac{m^3}{s}$$

$$\text{ANSWER: } 1.245 \times 10^5 \text{ Pa}$$

$$Q = A \bar{v}$$

$$\bar{v} = \frac{Q}{A}$$

$$\bar{v} = \frac{Q}{\pi r^2}$$

$$\bar{v} = \frac{4Q}{\pi d^2}$$

$$P_2 = P_a + \rho h$$

$$P_1 = P_2 + \frac{1}{2} \rho \left[\left(\frac{4Q}{\pi d_1^2} \right)^2 - \left(\frac{4Q}{\pi d_2^2} \right)^2 \right]$$

$$P_1 = 101325 \text{ Pa} + \frac{1}{2} \cdot 1000 \frac{\text{kg}}{\text{m}^3} \left[\left(\frac{4 \cdot 2.564 \times 10^{-4} \frac{\text{m}^3}{s}}{\pi (6.9 \times 10^{-3} \text{ m})^2} \right)^2 - \left(\frac{4 \cdot 2.564 \times 10^{-4} \frac{\text{m}^3}{s}}{\pi (0.02 \text{ m})^2} \right)^2 \right] = 1.245 \times 10^5 \text{ Pa}$$

5. A particular hydraulic system contains two fluid-filled cylinders which are connected by a tube. Cylinder 1 has a diameter of 2.1 cm and is capped with piston P_1 . Cylinder 2 has a diameter of 6.9 cm and is capped with piston P_2 . A force F_1 , equal to 80 N, is exerted on P_1 . What is the magnitude of the resulting force exerted by P_2 ?

$$P_1 = P_2$$

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$F_2 = \frac{A_2}{A_1} F_1$$

$$\frac{A_2}{A_1} = \frac{\pi r_2^2}{\pi r_1^2} = \frac{\pi d_2^2}{\pi d_1^2} = \left(\frac{d_2}{d_1} \right)^2$$

$$\text{ANSWER: } 8.637 \times 10^2 \text{ N}$$

$$F_2 = \left(\frac{d_2}{d_1} \right)^2 F_1 = \left(\frac{6.9 \text{ cm}}{2.1 \text{ cm}} \right)^2 \cdot 80 \text{ N} = 8.637 \times 10^2 \text{ N}$$

6. While Lauren is fishing, she begins to reel in a reel big fish. Her "FishFinder 3000" tells her that this particular fish has a volume of 7600 cm³. As she's reeling the fish in, she notes that her reel ($I = MR^2/2$) begins at rest and revolves 6 times in 6 seconds. The 390-g reel has a radius of 4 cm and a friction torque of 18 N cm. How much force did Lauren apply to the reel's crank handle to reel the fish in if the handle is a distance of 5.2 cm from the center of the reel and the fish was pulled vertically upwards through freshwater, which has a density of 1000 kg/m³? Later measurements of the fish reveals it had a mass of 8.44 kg. You may ignore the effects of the drag force.

T_h = handle torque
 F_t = tension in line
 T_f = friction torque

$$\Sigma \tau = I \alpha$$

$$T_h - T_f = I \alpha$$

$$F_h r_h = I \alpha + T_f r + T_f$$

for fish:

$$\Sigma F_y = m a$$

$$F_t + F_b - w = m a$$

$$F_t + \rho V g - m g = m a$$

$$F_t = \rho(m - \rho V) + m \frac{2r \Delta \theta}{t^2}$$

$$\Delta \theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\alpha = \frac{2 \Delta \theta}{t^2} \quad \because \omega_0 = 0$$

$$a = r \alpha = \frac{2r \Delta \theta}{t^2}$$

$$\Delta \theta = 6 \text{ rev} = 12 \pi \text{ rad}$$

$$\text{ANSWER: } 10.350 \text{ N}$$

$$V = 7600 \text{ cm}^3 \left(\frac{1 \text{ m}^3}{10^6 \text{ cm}^3} \right) = 7.6 \times 10^{-3} \text{ m}^3$$

$$F_h r_h = \frac{M r^2}{2} \cdot \frac{2 \Delta \theta}{t^2} + \left[\rho(m - \rho V) + m \frac{2r \Delta \theta}{t^2} \right] r + T_f$$

$$F_h = \frac{M r^2 \Delta \theta}{t^2} + \frac{2 m r^2 \Delta \theta}{t^2} + r \rho(m - \rho V) + T_f$$

$$F_h = \frac{0.39 \text{ kg} \cdot 0.04 \text{ m}^2 \cdot 12 \pi}{6^2 \text{ s}^2} + \frac{2 \cdot 8.44 \text{ kg} \cdot 0.04 \text{ m}^2 \cdot 12 \pi}{6^2 \text{ s}^2} + 0.04 \text{ m} \cdot 9.8 \frac{\text{m}}{\text{s}^2} (8.44 \text{ kg} - 1000 \frac{\text{kg}}{\text{m}^3} \cdot 7.6 \times 10^{-3} \text{ m}^3) + 18 \times 10^{-2} \text{ N m}$$

$$F_h = 10.350 \text{ N}$$

7. Hannah attaches a nozzle of diameter 6.6 mm to a garden hose of diameter 2.5 cm. If the hose fills up a 2-L bucket in 10.5 seconds, what is the speed of the water that leaves the nozzle?

$$Q = \frac{2\text{ L}}{10.5\text{ s}} \left| \frac{\text{m}^3}{1000\text{ L}} \right| = 1.905 \times 10^{-4} \frac{\text{m}^3}{\text{s}}$$

ANSWER: $\underline{5.568 \frac{\text{m}}{\text{s}}}$ ✓

$$Q = A\bar{v}$$

$$\bar{v} = \frac{Q}{A}$$

$$\bar{v} = \frac{Q}{\pi r^2}$$

$$\bar{v} = \frac{4Q}{\pi d^2} = \frac{4 \cdot 1.905 \times 10^{-4} \frac{\text{m}^3}{\text{s}}}{\pi \cdot 0.0066^2 \text{ m}^2} = 5.568 \frac{\text{m}}{\text{s}}$$

8. A freshwater reservoir has a surface area of 59 km² and an average depth of 27 m. What mass of water is held behind the dam?

$$\rho = \frac{m}{V}$$

$$59 \text{ km}^2 \left| \frac{(10^3)^2 \text{ m}^2}{\text{km}^2} \right| = 59 \times 10^6 \text{ m}^2$$

ANSWER: $\underline{1.593 \times 10^{12} \text{ kg}}$ ✓

$$m = \rho V$$

$$m = \rho A h$$

$$m = 1000 \frac{\text{kg}}{\text{m}^3} \cdot 59 \times 10^6 \text{ m}^2 \cdot 27 \text{ m} = 1.593 \times 10^{12} \text{ kg}$$

9. Intravenous (IV) infusions are usually made with the help of the gravitational force. Assuming that the density of the fluid being administered is 1.00 g/mL, at what height should the IV bag be placed above the entry point so that the fluid only just enters the vein if the blood pressure in the vein is 5.8 mmHg above atmospheric pressure? Recall that 1.0 mmHg = 133.32 Pa.

$$P_f = P_b \longrightarrow P_{\text{atm}} + P_g = P_{\text{atm}} + P_{\text{b above}}$$

ANSWER: $\underline{7.89 \times 10^{-2} \text{ m}}$ ✓

$$P_{\text{b above}} = \rho g h$$

$$h = \frac{P_{\text{b above}}}{\rho g}$$

$$P_{\text{b above}} = 5.8 \text{ mmHg} \left| \frac{133.32 \text{ Pa}}{1 \text{ mmHg}} \right| = 7.7326 \times 10^2 \text{ Pa}$$

$$\rho = 1 \frac{\text{g}}{\text{mL}} \left| \frac{\text{kg}}{1000\text{ g}} \right| \left| \frac{1000\text{ mL}}{\text{L}} \right| \left| \frac{1000\text{ L}}{\text{m}^3} \right| = 1000 \frac{\text{kg}}{\text{m}^3}$$

$$h = \frac{7.7326 \times 10^2 \text{ Pa}}{1000 \frac{\text{kg}}{\text{m}^3} \cdot 9.8 \frac{\text{m}}{\text{s}^2}} = 7.89 \times 10^{-2} \text{ m}$$

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1. $4.02 \times 10^{10} \text{ N}$
2. $1.55 \times 10^{10} \text{ capillaries}$
3. 1.52 g/cm^3
4. 124.5 kPa
5. 863.67 N
6. 10.35 N
7. 5.57 m/s
8. $1.59 \times 10^{12} \text{ kg}$
9. 7.87 cm